Radiative corrections for exclusive π^0 electroproduction at CLAS12: rad/Born Monte Carlo against the EXCLURAD formula

Status report — statistics are not yet sufficient for a definite conclusion

exclurad_py study, branch `rc-method-comparison`

21 October 2026

Abstract

We compare two ways of correcting exclusive $ep \to e'p'\pi^0$ yields for radiation. (B) The method of the CLAS12 analysis note (Korover *et al.*, Eq. 20): generate radiative and non-radiative Monte Carlo, pass both through GEMC and the same cuts, and take $\delta_{\rm RC} = N_{\rm rad}/N_{\rm Born}$ bin by bin. (A) The EXCLURAD way: compute $\delta = \sigma_{\rm obs}/\sigma_{\rm Born}$ with the same radiative engine (`exclurad_py`, `rc_factor`), where the whole selection is represented by one number, $v_{\rm cut}$. Both use the model `pi0.amp2609`. With the analysis note's exclusivity cuts, a single $v_{\rm cut} \approx 0.06$ GeV^2 reproduces the ϕ-integrated rad/Born correction in every combination of Q^2, x_B, $-t$ bins within 2–10% errors; $v_{\rm cut} = 0.2$ is too loose by 5–10%. The equivalent $v_{\rm cut}$ follows the exclusivity windows, not the kinematics. **These are ϕ-integrated results.** The ϕ dependence of the correction, which decides the separation of $\sigma_T + \varepsilon\sigma_L$, σ_{TT} and σ_{LT}, is not yet measured with useful precision: the present sample (1M rad + 1M Born events per polarity) is too small. A larger sample is being generated.

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1 The two methods

B — rad/Born Monte Carlo (analysis note, Eq. 20).

$$\delta_B(\phi) = \frac{N^{\rm rec}_{\rm rad}(\phi)\,\sigma_{\rm rad}/N^{\rm gen}_{\rm rad}}{N^{\rm rec}_{\rm Born}(\phi)\,\sigma_{\rm Born}/N^{\rm gen}_{\rm Born}}, \tag{1}$$

both arms reconstructed and passed through the same cuts. No acceptance is divided out: the note applies the Born acceptance separately, so δ_B carries the ratio of acceptances inside it. Exactly,

$$\delta_B = \eta_{\rm gen}(\text{no cut}) \times \frac{\mathrm{Acc}_{\rm rad}}{\mathrm{Acc}_{\rm Born}}, \tag{2}$$

and the exclusivity cuts act only through the acceptance ratio.

A — the EXCLURAD formula. $\eta = \sigma_{\rm obs}(v_{\rm cut})/\sigma_{\rm Born}$ from `rc_factor`, at observed kinematics. The only exclusivity cut `rc_factor` knows is $v_{\rm cut}$: $MM^2(e'p') < v_{\rm cut} + m^2_{\pi^0}$. It knows nothing of missing energy, $MM^2(e'p'\pi^0)$, MP_T or coplanarity. For any binning, A per bin is evaluated per event and averaged as

$$A_{\rm bin} = N \Big/ \sum_i 1/\eta(x_i) \tag{3}$$

over the accepted radiative events, which equals the Born- and acceptance-weighted mean of η without using the Born Monte Carlo.

ϕ. The RC engine defines ϕ for the proton-detected topology from the recoil proton, $\phi = \phi_{p'} + 180^\circ$ (Trento). The analysis must use the same definition; all results below do. With ϕ taken from the $\gamma\gamma$ pair instead, the generator-level closure fails for loose cuts (the radiative tail lands at 180° instead of 0°).

Engine. `exclurad_py`, `exclurad_py/core/rc.py`, from the branch checkout. Values at the campaign commit (`2470395`) and at the current branch are identical for $v_{\rm cut} \le 0.3$; they differ by 0.3–0.8% with no cut. The Fortran EXCLURAD cannot run these models and is not used.

2 Monte Carlo sample and its limits

- Campaign `/volatile/clas12/vpk/prod_pi0_rga_t25_20260928`: RG-A fall 2018, 10.6 GeV, inbending and outbending, 50 nA background; 1M rad and 1M Born (990k Born inbending) per polarity. Box Q^2 0.8–10, W^2 3.24–16, t $-2.5 \ldots -0.001$, generated with $v_{\rm cut} = 0$.
- Generator commits all contain the radiative photon-pair fix (`2a1c904`) and the stage-2 fix (`225d4d9`).
- **Statistics.** After the full note selection: 35k Born and 23k rad events in all kinematics.
- **`itarg=1` in the Born arm too**: 2.8% of Born events in a typical bin carry pre-vertex radiation; this biases B by about +3% against A.
- Model `amp2609` (not `amp2614`); proton target only.
- The data in all tests are the radiative MC itself (or half of it): data and correction come from the same generator and model. The test is therefore of the *method*, not of the model.
- The ntuple carries no vertex and no sector: the vertex cuts and the same-sector veto of the note are not applied (they act identically on both arms).

3 Event selection: every cut variable

The note's Sec. 3.5 cuts: $Q^2 > 1$, $W > 2$, $p_{e'} > 2$; $|ME| < 0.2$ GeV, $|MM^2(e'p'\pi^0)| < 0.004$ GeV^2, $MP_T < 0.75$ GeV, coplanarity $\theta(X, \pi^0) < 4°$, where X is the missing particle of $ep \to e'p'X$ and π^0 the reconstructed $\gamma\gamma$ pair; π^0 identification by $M_{\gamma\gamma}$ (3σ of the MC) and opening angle $> 1°$. These are data windows after the note's momentum corrections; our uncorrected GEMC is wider, and the literal windows keep 39–43% of Born. They act on both arms alike, which is all B needs.

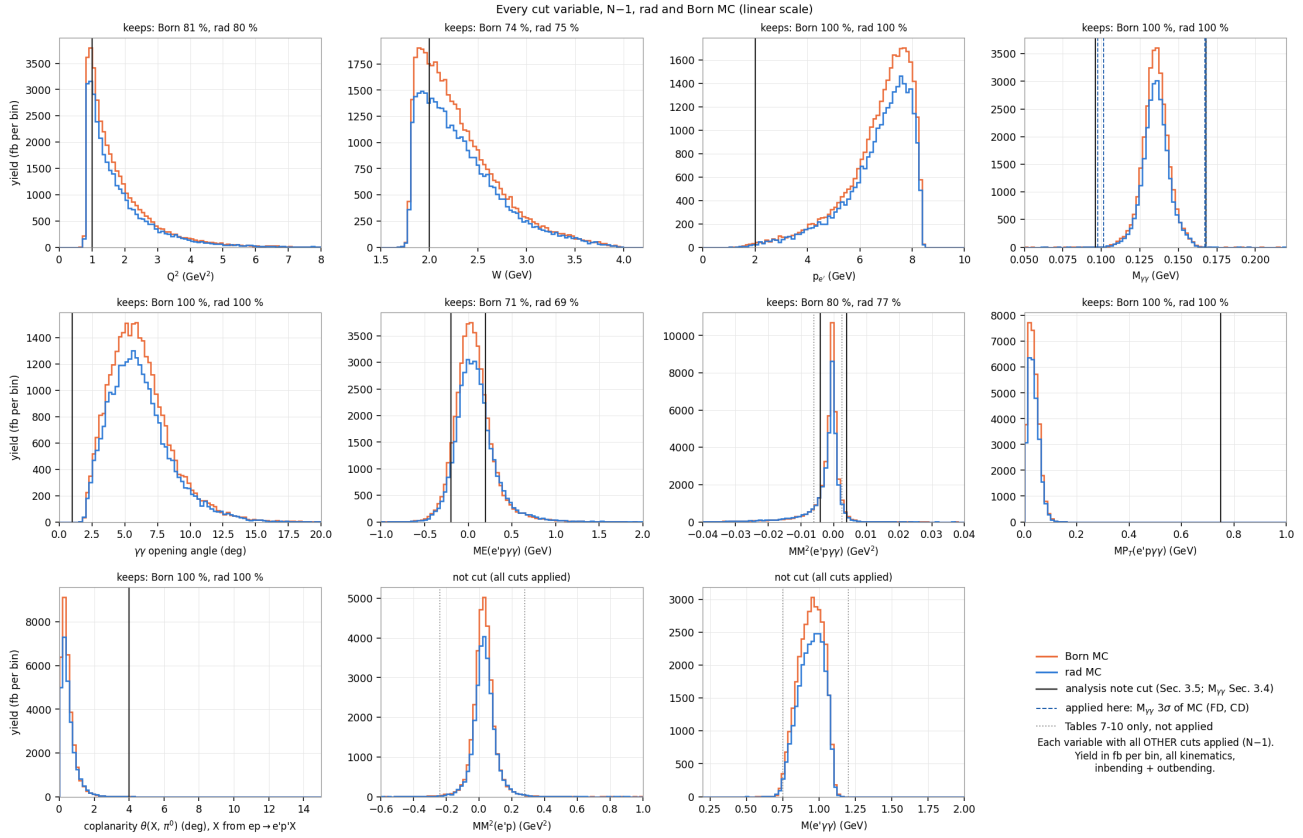


Figure 1: Every cut variable with all *other* cuts applied (N−1), rad and Born MC, all kinematics. Solid: the note's cut. Dotted: windows of Tables 7–10 that Sec. 3.5 does not list.

cut, given all others	Born kept	rad kept
$Q^2 > 1$	81%	80%
$W > 2$	74%	75%
$\|ME\| < 0.2$	**71%**	**69%**
$\|MM^2(e'p'\pi^0)\| < 0.004$	**80%**	**77%**
$p_{e'}$, $M_{\gamma\gamma}$, opening, MP_T, coplanarity	100%	100%

Table 1: Only ME and $MM^2(e'p'\pi^0)$ treat rad and Born differently.

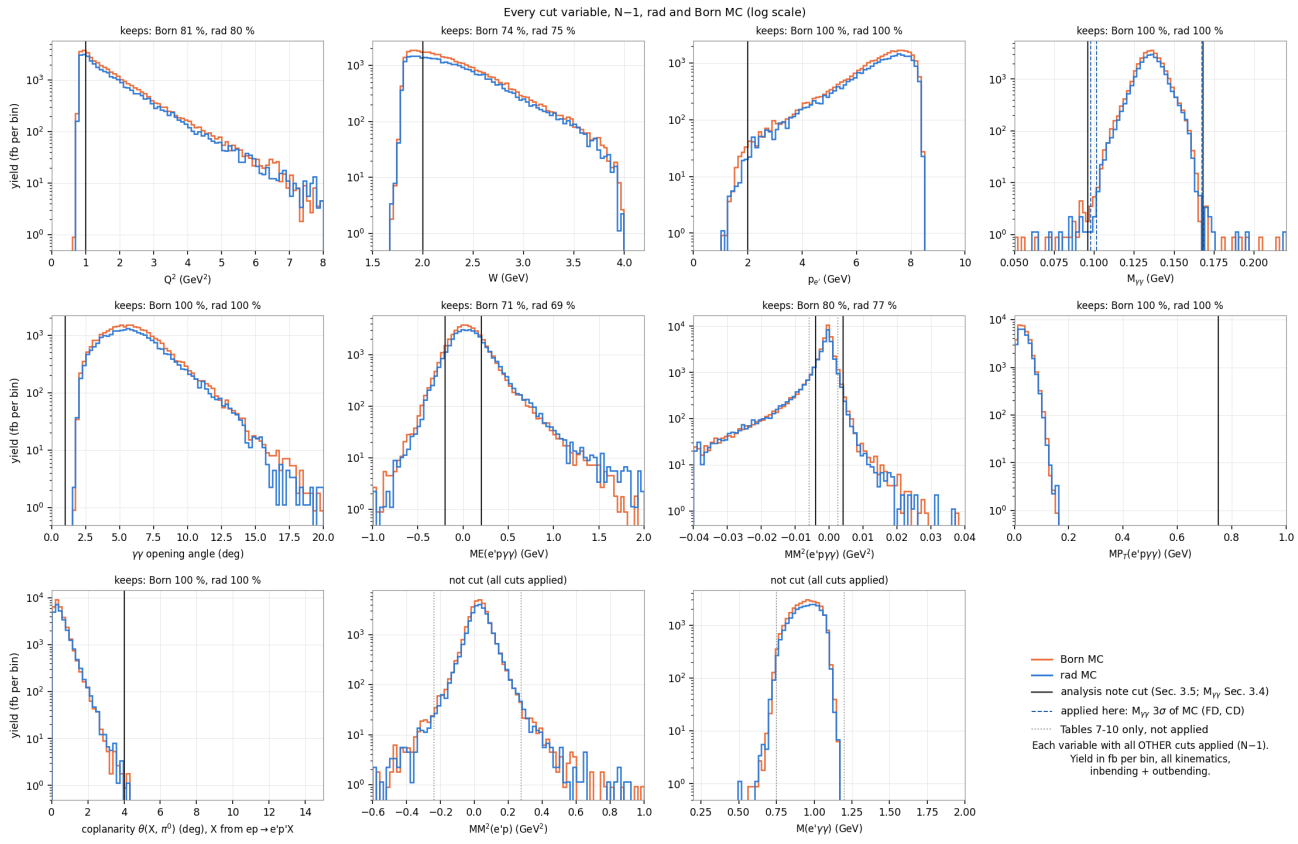


Figure 2: As above, logarithmic scale.

4 How the exclusivity cuts act on radiation

For a radiative event with an unseen photon k:

$$v = MM^2(e'p') - m_\pi^2 = 2p_\pi \cdot k = 2E_k E_\pi (1 - \beta_\pi \cos\theta_{\pi k}), \tag{4}$$

$$ME = E_k, \qquad MM^2(e'p'\pi^0) = k^2 = 0, \qquad MP_T = k_T, \tag{5}$$

$$\tan\theta(X, \pi^0) \approx \frac{E_k \sin\theta_{\pi k}}{E_\pi + E_k \cos\theta_{\pi k}}. \tag{6}$$

The missing-energy cut cuts the photon energy and maps onto v through the photon's angle to the π^0 (median 21°). $MM^2(e'p'\pi^0)$ vanishes for any real photon: it cuts radiation only through a resolution that grows with E_k. The coplanarity angle grows like $\sin\theta_{\pi k}$ while v grows like $\theta_{\pi k}^2$, so it is the weaker cut.

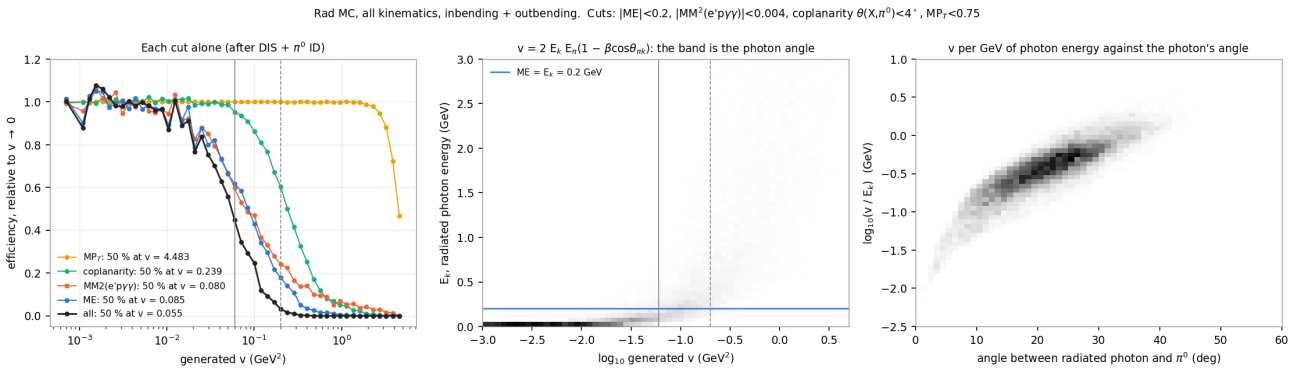


Figure 3: Left: efficiency of each cut alone against the generated v; 50% points ME 0.085, $MM^2(e'p'\pi^0)$ 0.080, coplanarity 0.24, MP_T 4.5, all together 0.055 GeV^2. Middle: photon energy against v. Right: v/E_k against the photon–π^0 angle.

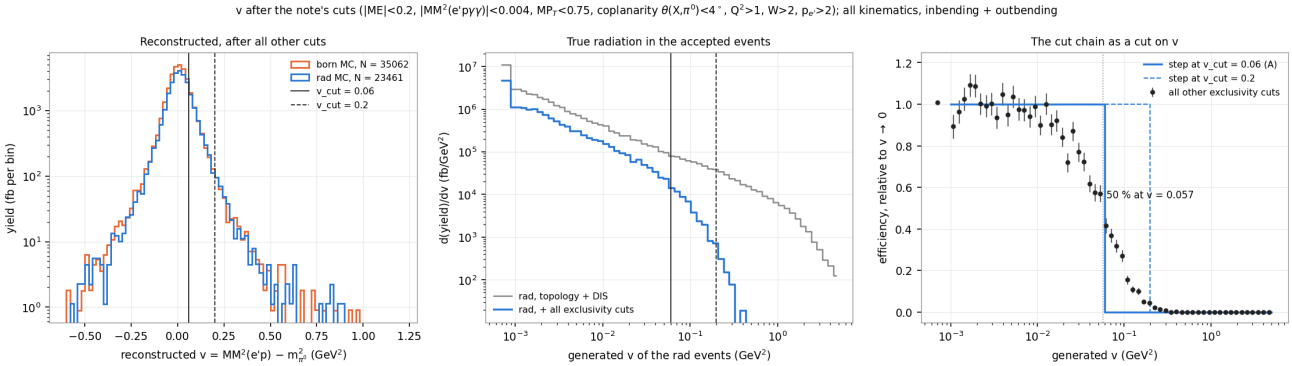


Figure 4: v after all other cuts. Left: reconstructed. Middle: generated v of the accepted rad events before and after the exclusivity cuts. Right: the cut chain as an efficiency in v: flat to $v \approx 0.01$, 50% at $v = 0.057$, near zero above 0.2. Only 2.0% of the accepted rad yield has $v > 0.066$, 0.09% has $v > 0.2$.

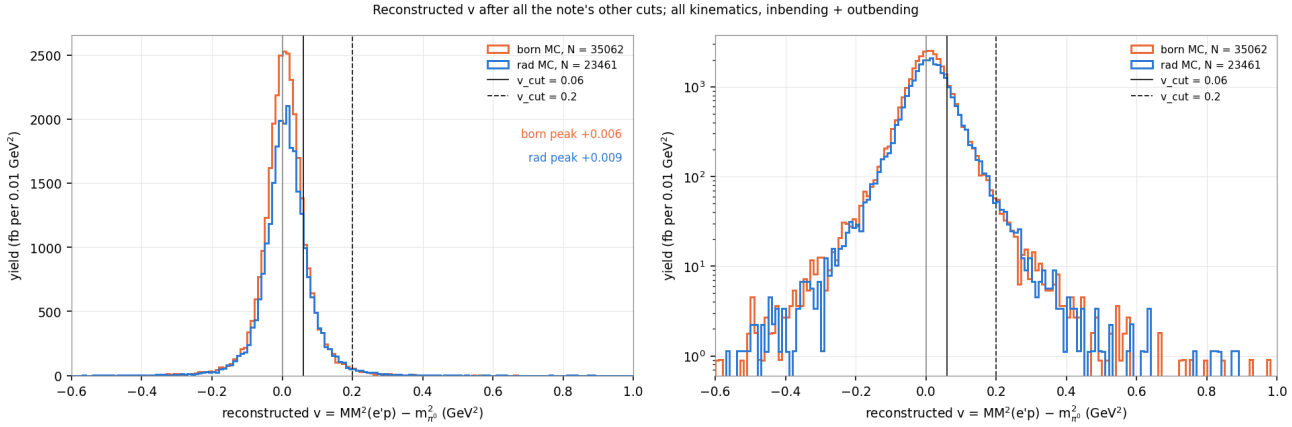


Figure 5: Reconstructed v after all other cuts, linear and log. The peak is at +0.006 (Born) and +0.009 GeV^2 (rad): reconstructed momenta are 0.6% (proton) and 0.1% (electron) low, no energy-loss correction. The core half-width is ≈ 0.04 GeV^2; the reconstructed $MM^2(e'p')$ cannot resolve a $v_{\rm cut}$ of 0.06.

5 One wide bin, ϕ by ϕ (MC-calibrated windows)

First pass, $1.5 < Q^2 < 2.5$, $0.15 < x_B < 0.30$, $0.2 < -t < 0.6$, with each exclusivity window set to $\pm 3\sigma$ of the Born MC (not the note's literal values).

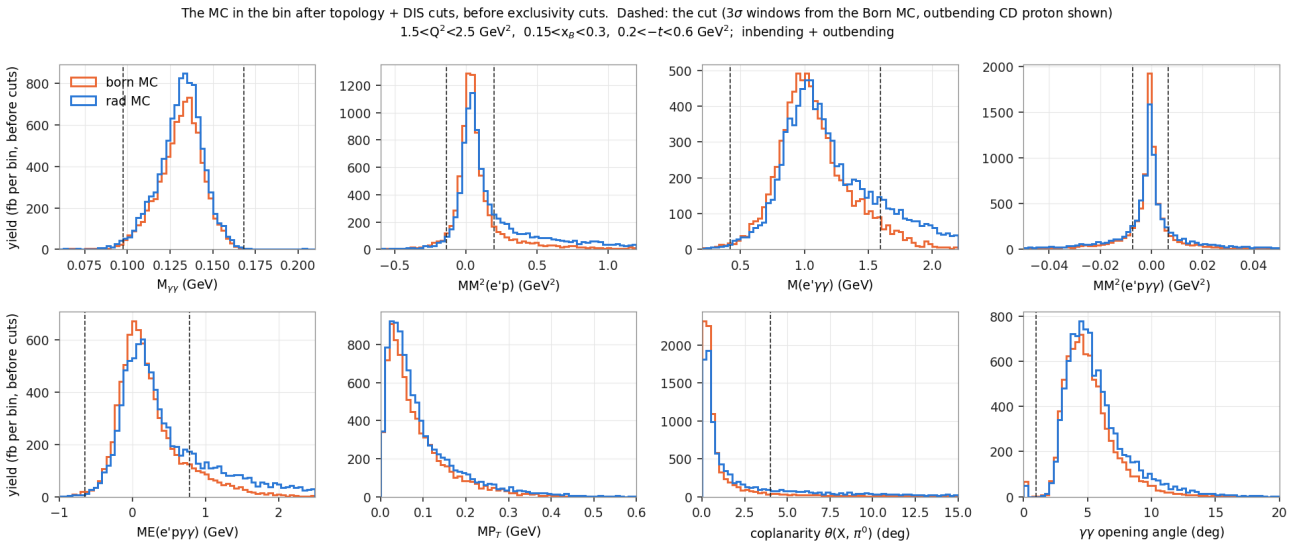


Figure 6: The MC in the bin before exclusivity cuts.

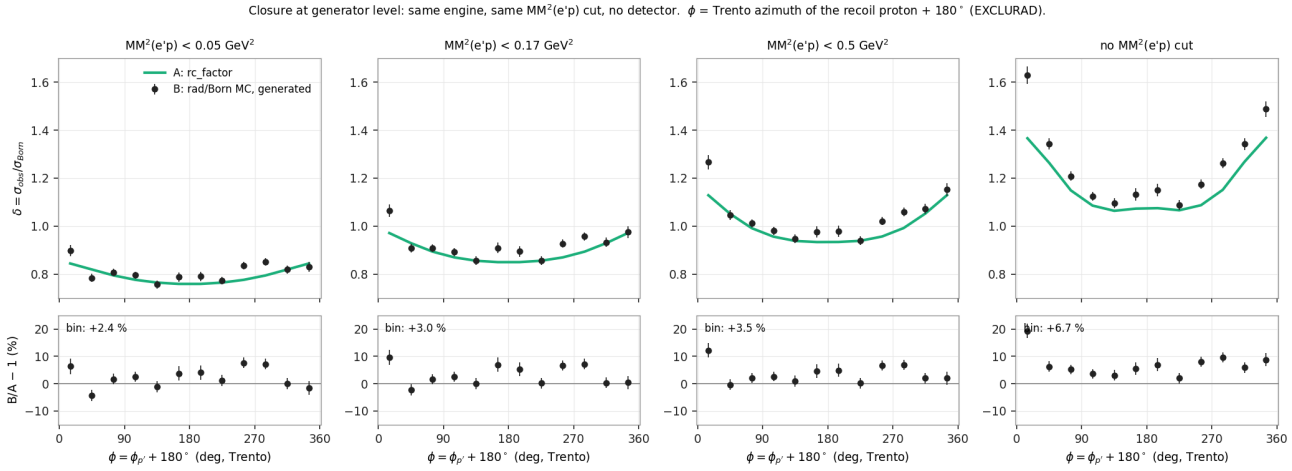


Figure 7: Closure at generator level: same engine, same $MM^2(e'p')$ cut, no detector. B follows A in shape at every cut; the bin-integrated +3% is `itarg` in the Born arm.

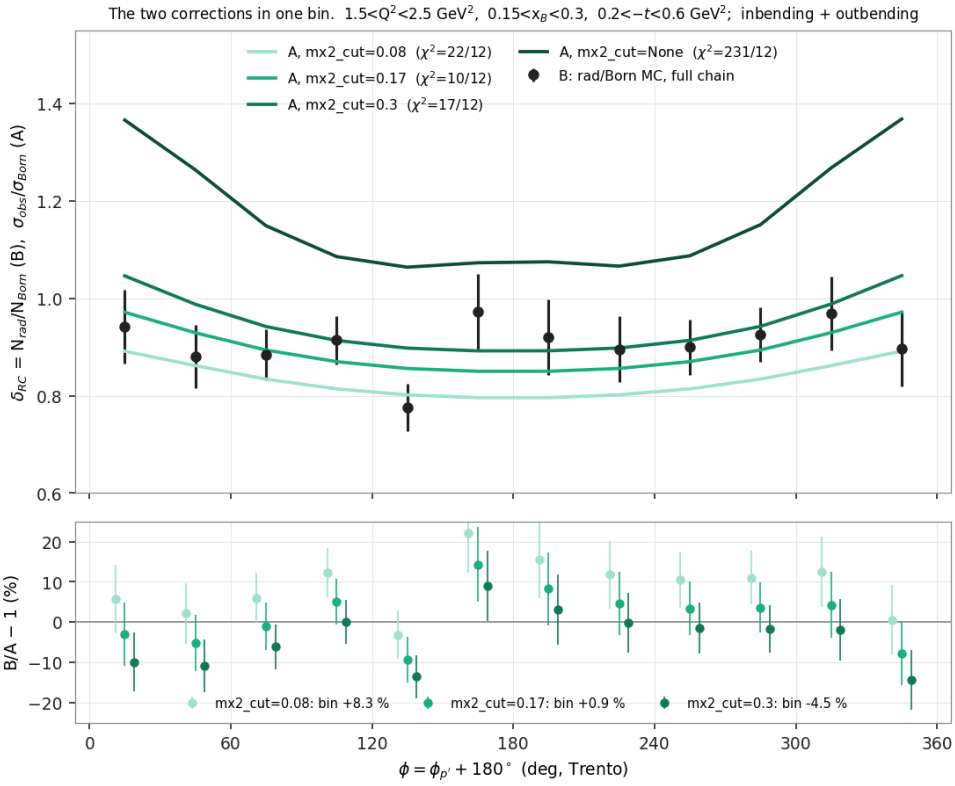


Figure 8: $\delta_B(\phi)$ with the full chain against $\delta_A(\phi)$ for several $v_{\rm cut}$; B/A-1 in % below. Best $v_{\rm cut} = 0.17$, $\chi^2 = 9.7/12$; errors 5–8% per ϕ bin.

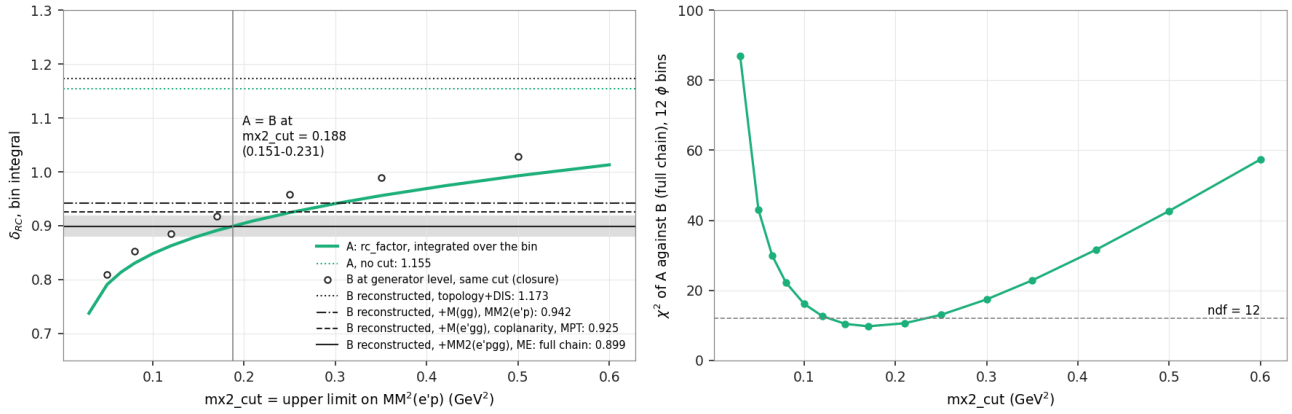


Figure 9: Bin integral against $v_{\rm cut}$ and χ^2 of A against B.

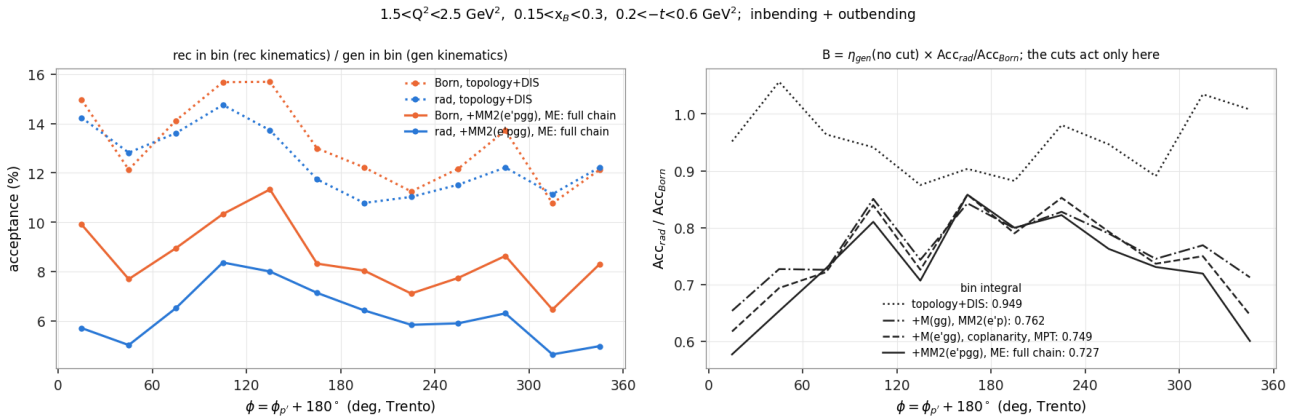


Figure 10: The two acceptances and their ratio: the cuts act on δ_B only through $\mathrm{Acc_{rad}}/\mathrm{Acc_{Born}}$ (0.95 after topology+DIS, 0.73 after the full chain).

6 The note's grid, MC-calibrated windows

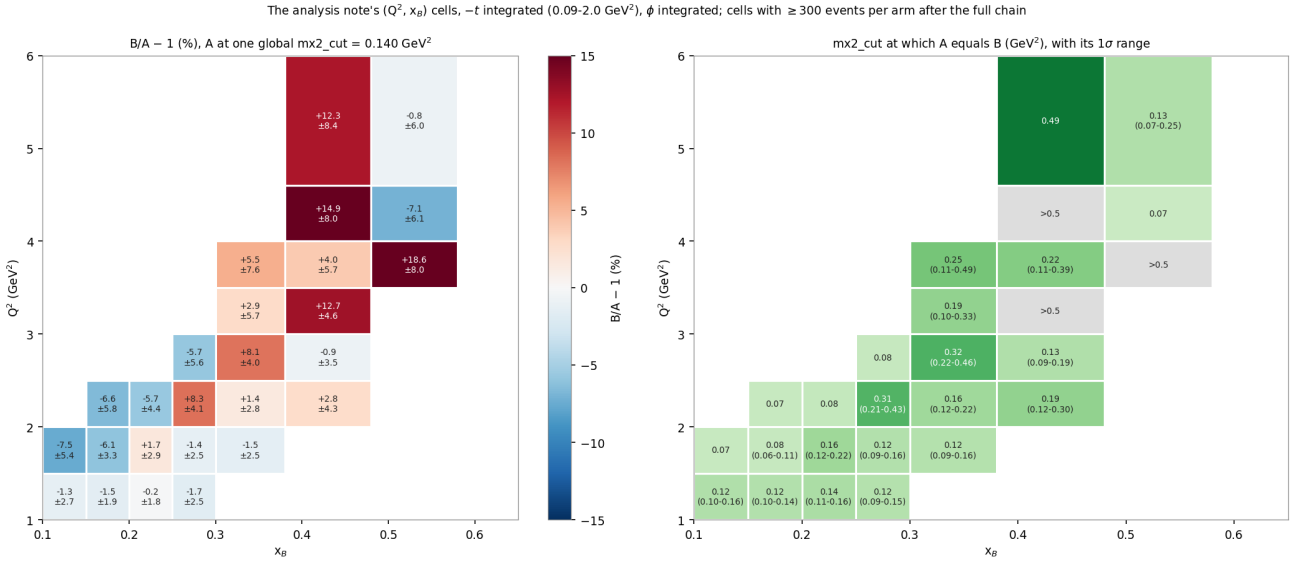


Figure 11: (Q^2, x_B) cells, $-t$ and ϕ integrated. Left: B/A-1 at the best global $v_{\mathrm{cut}} = 0.14$ ($\chi^2 = 42/26$). Right: the v_{cut} at which A equals B, cell by cell.

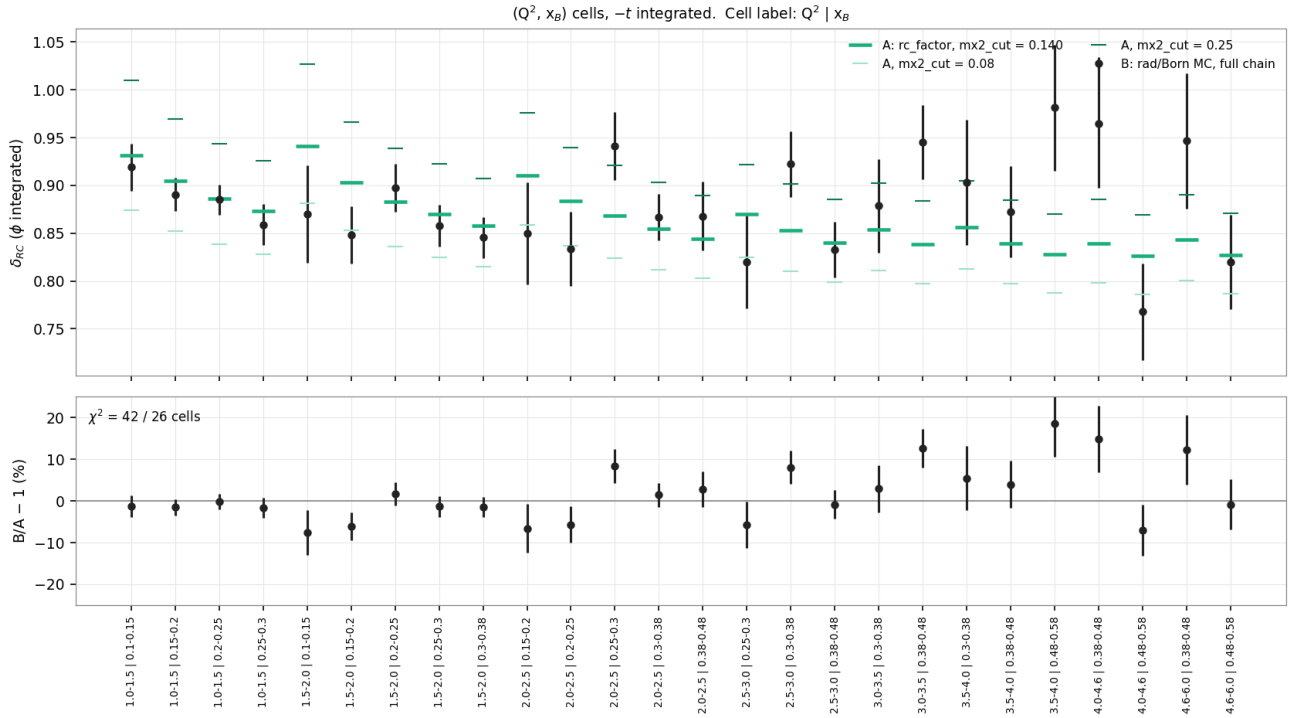


Figure 12: B and A per (Q^2, x_B) cell, B/A−1 in % below. With these wider windows the high Q^2/x_B cells are 10–18% off.

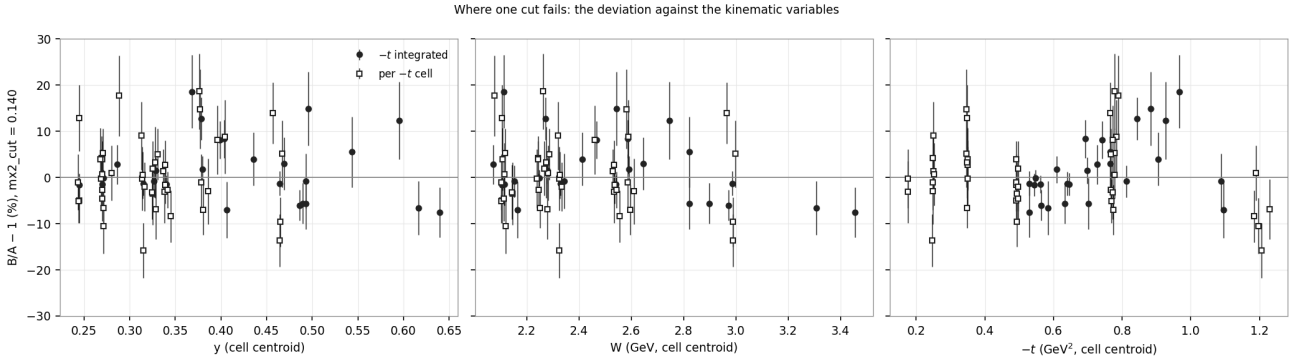


Figure 13: Deviation against y, W and $-t$ of the cell centroid.

7 The note's cuts as printed: every combination of Q^2, x_B, $-t$

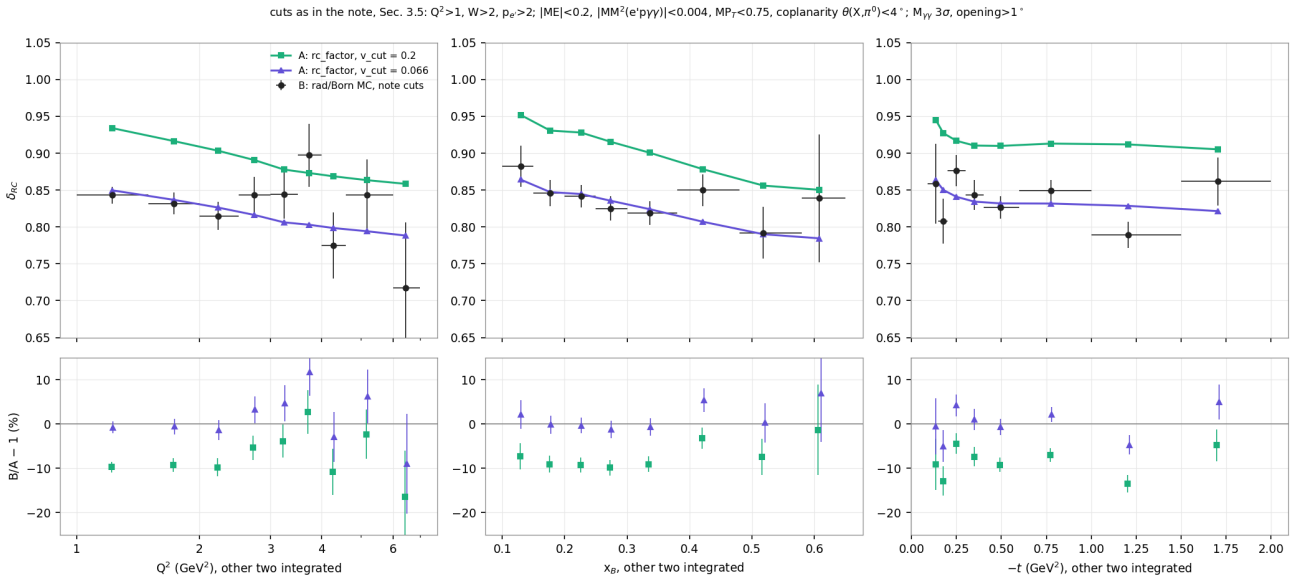


Figure 14: One variable at a time, the other two integrated. A at $v_{\rm cut} = 0.2$ (green) is 5–10% above B; A at the best global $v_{\rm cut} = 0.066$ (violet) agrees.

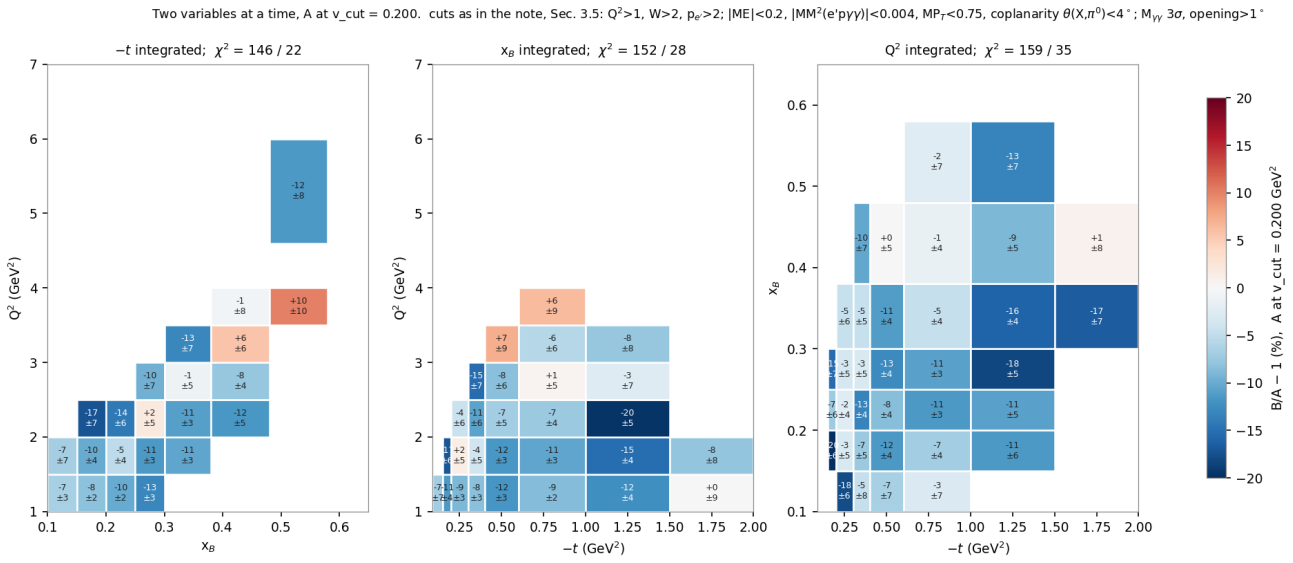


Figure 15: Two variables at a time, A at $v_{\rm cut} = 0.2$. Cell text: B/A−1 (%) ± error.

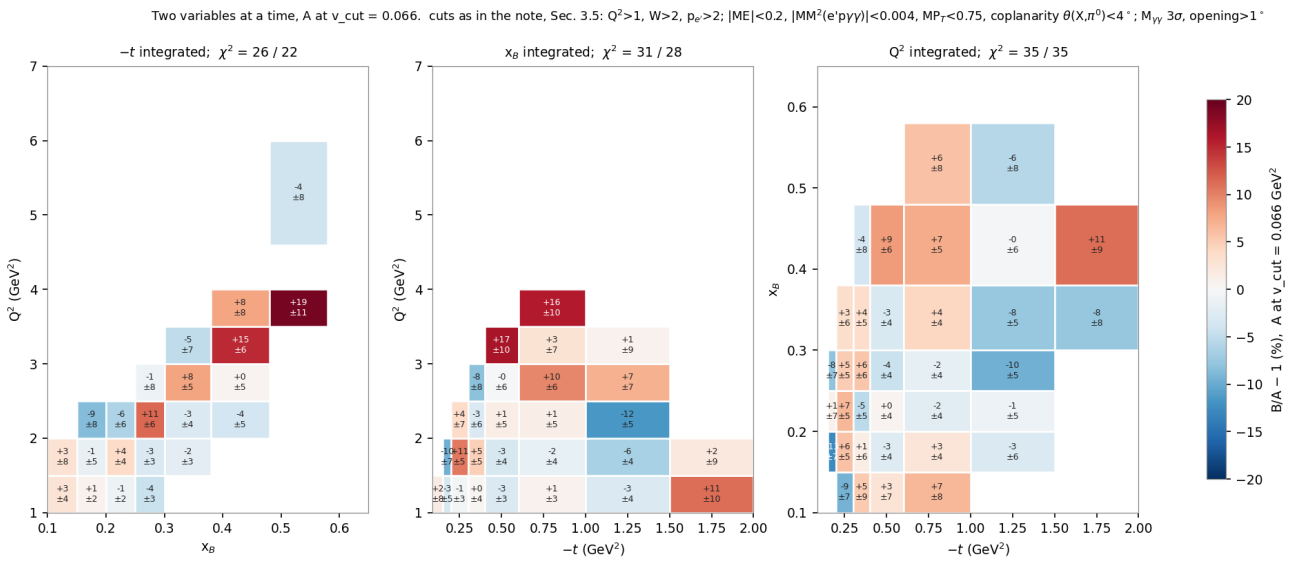


Figure 16: Two variables at a time, A at $v_{\text{cut}} = 0.066$.

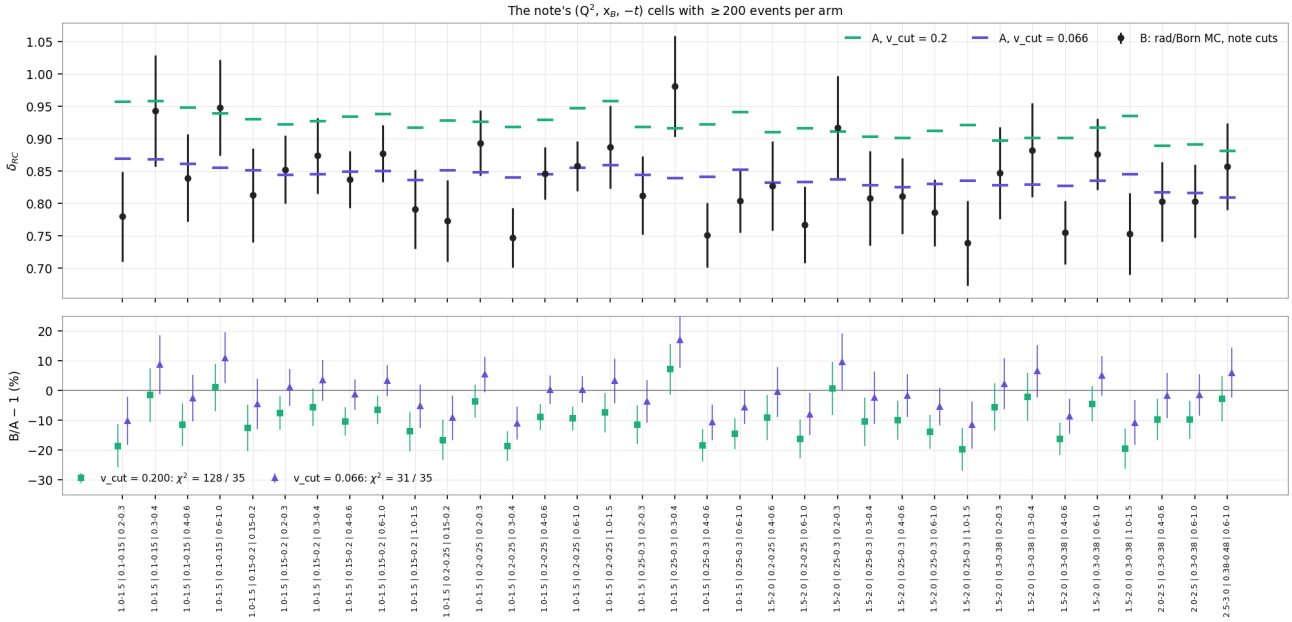


Figure 17: All three variables, cells with ≥ 200 events per arm.

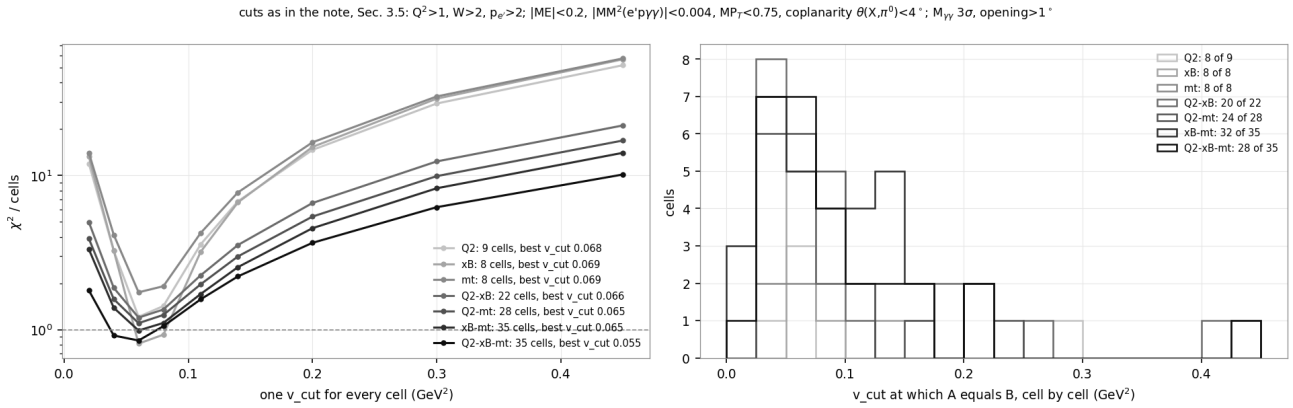


Figure 18: χ^2/cells of one global $v_{\rm cut}$ for every binning, and the cell-by-cell equivalent $v_{\rm cut}$.

binning (rest integrated)	cells	best $v_{\rm cut}$	χ^2 at 0.06
Q^2	9	0.068	11
x_B	8	0.069	6
$-t$	8	0.069	14
$Q^2 \times x_B$	22	0.066	26
$Q^2 \times (-t)$	28	0.065	31
$x_B \times (-t)$	35	0.065	35
$Q^2 \times x_B \times (-t)$	35	0.055	30

Table 2: ϕ-integrated: one $v_{\rm cut} \approx 0.06$–0.07 fits every combination with $\chi^2/{\rm ndf} \approx 1$ at the present 2–10% errors.

8 ϕ dependence and the structure functions

The final goal is the separation $d\sigma/d\phi \propto \sigma_U + \varepsilon\sigma_{TT}\cos 2\phi + \sqrt{2\varepsilon(1+\varepsilon)}\,\sigma_{LT}\cos\phi$, $\sigma_U = \sigma_T + \varepsilon\sigma_L$, so the test that matters is not the integral of δ but what each correction does to the three structure functions. In two wide bins:

- pseudo-data: the reconstructed rad MC after the note's cuts, *even* event numbers;
- δ_B from the *odd* rad events and the Born MC (so B is not the data divided by itself);
- δ_A from `rc_factor` per event, $v_{\rm cut} = 0.06$ and 0.2;
- $\sigma(\phi) = N_{\rm data}(\phi)/({\rm Acc}_{\rm Born}(\phi)\,\delta(\phi))$, with ${\rm Acc}_{\rm Born}$ = Born reconstructed / Born generated in the bin;
- fitted for σ_U, σ_{TT}, σ_{LT} with bin-averaged cosines, and compared with the same fit of the generated Born distribution (the model).

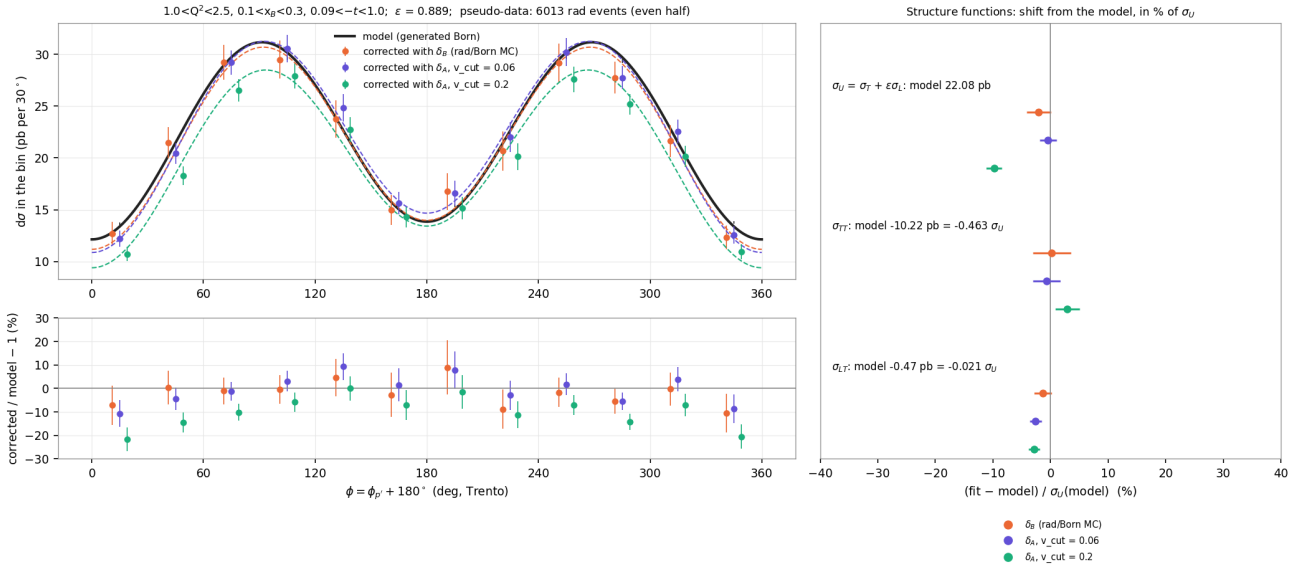


Figure 19: Bin $1.0 < Q^2 < 2.5$, $0.10 < x_B < 0.30$, $0.09 < -t < 1.0$, $\varepsilon = 0.889$, 6013 pseudo-data events. Left: corrected ϕ distributions and fits against the model; corrected/model−1 in % below. Right: shift of each structure function from the model, in % of σ_U.

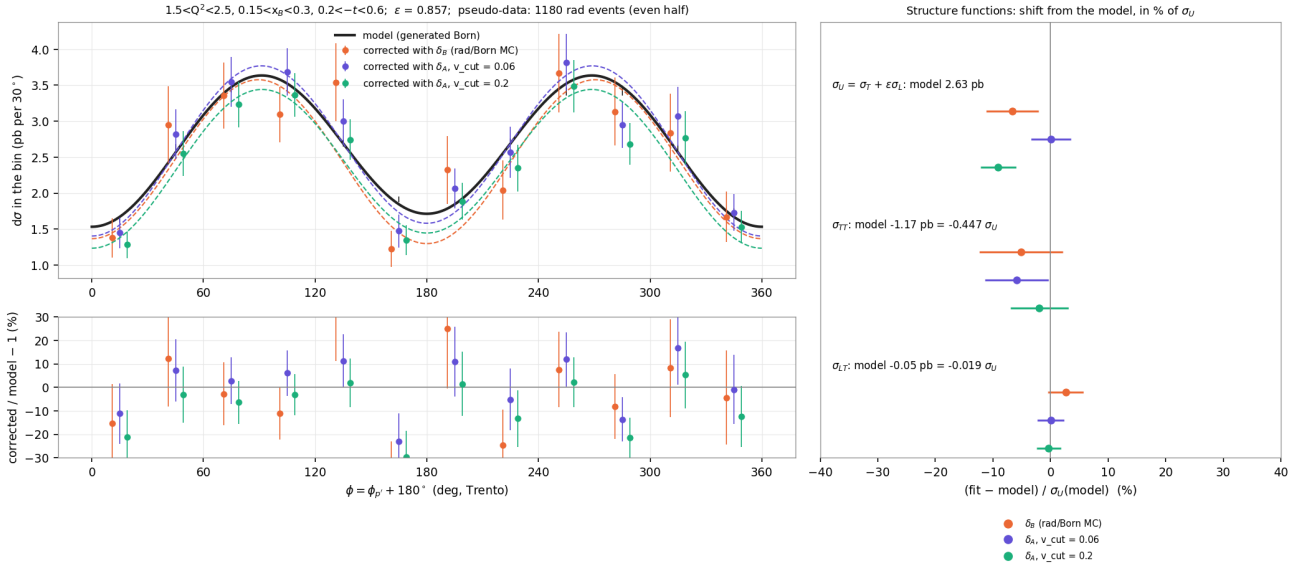


Figure 20: Bin $1.5 < Q^2 < 2.5$, $0.15 < x_B < 0.30$, $0.2 < -t < 0.6$, $\varepsilon = 0.857$: only 1180 pseudo-data events.

bin	correction	$\sigma_U/\sigma_U^{\rm model}$	$(\sigma_{TT}-\sigma_{TT}^{\rm model})/\sigma_U$	$(\sigma_{LT}-\sigma_{LT}^{\rm model})/\sigma_U$
wide2	B (rad/Born)	0.979 ± 0.021	$+0.002 \pm 0.033$	-0.013 ± 0.015
(6013 ev.)	A, $v_{\rm cut} = 0.06$	0.996 ± 0.015	-0.007 ± 0.024	-0.026 ± 0.011
	A, $v_{\rm cut} = 0.2$	0.902 ± 0.014	$+0.029 \pm 0.021$	-0.029 ± 0.010
wide1	B	0.934 ± 0.046	-0.051 ± 0.073	$+0.027 \pm 0.031$
(1180 ev.)	A, $v_{\rm cut} = 0.06$	1.001 ± 0.034	-0.059 ± 0.055	$+0.000 \pm 0.024$
	A, $v_{\rm cut} = 0.2$	0.909 ± 0.031	-0.019 ± 0.050	-0.003 ± 0.021

Table 3: Model values: wide2 $\sigma_{TT}/\sigma_U = -0.463$, $\sigma_{LT}/\sigma_U = -0.021$; wide1 -0.447 and -0.019.

What this shows, at this statistics:

- $v_{\rm cut} = 0.2$ underestimates σ_U by 10% (7σ in wide2); $v_{\rm cut} = 0.06$ and B recover it.
- The errors on σ_{TT} are 2–3% of σ_U (5–7% of σ_{TT}) in the larger bin, 5–7% of σ_U in the smaller. Differences between the corrections are within them.
- σ_{LT} is small here ($\approx -0.02\,\sigma_U$) and its error, 0.010–$0.015\,\sigma_U$, is as large as σ_{LT} itself. The A-corrected σ_{LT} sits $0.026 \pm 0.011\,\sigma_U$ below the model (2.4σ), B 0.013 ± 0.015: suggestive, not conclusive.
- The data error dominates; the B correction adds its own statistical error, which A does not.

The ϕ dependence of the correction, and with it the effect on σ_{TT} and σ_{LT}, cannot be settled with the present sample.

9 Conclusions so far

1. **The engine is consistent with itself.** At generator level, with the same $MM^2(e'p')$ cut, A and B agree in shape; the residual +3% is `itarg` in the Born arm.
2. **ϕ must be defined as in EXCLURAD**, $\phi_{p'} + 180^\circ$, in the analysis and in the RC alike.
3. **Only ME and $MM^2(e'p'\pi^0)$ cut radiation**; MP_T and coplanarity add nothing once they are applied. Together they act as a soft cut on the true v with its 50% point at $v = 0.057$.
4. **ϕ-integrated**, one $v_{\rm cut} \approx 0.06$–0.07 GeV^2 reproduces B in every combination of Q^2, x_B, $-t$ within 2–10%; $v_{\rm cut} = 0.2$ is 5–10% off. The equivalent $v_{\rm cut}$ is a property of the exclusivity windows and the resolution (0.14 with wider windows), and must be derived for each cut chain.
5. **No definite conclusion is possible yet on the ϕ dependence.** An integrated agreement says nothing about the $\cos\phi$ and $\cos 2\phi$ terms of δ. A difference $\delta_B/\delta_A = 1 + a\cos\phi + b\cos 2\phi$ shifts σ_{LT} by $\approx a\,\sigma_U/\sqrt{2\varepsilon(1+\varepsilon)}$ and σ_{TT} by $\approx b\,\sigma_U/\varepsilon$; for π^0, where σ_{TT} and σ_{LT} are a fraction of σ_U, a 2–3% term becomes a 10–30% change of the structure function. The separation of $\sigma_T + \varepsilon\sigma_L$, σ_{TT}, σ_{LT} is the goal of the experiment, so this is the test that matters, and the present statistics cannot make it.

10 What is needed

- **Statistics**: $\sim 10^4$ accepted events per arm per bin to measure the $\cos\phi$ and $\cos 2\phi$ terms of δ to $\approx 1\%$; roughly 10–20M generated per arm and polarity for a few wide bins, more for the note's grid.

- Born arm with `itarg`=0.
- Pseudo-data from a *different* model (`amp2614`) than the correction (`amp2609` twins), so the test is not circular.
- Vertex and sector in the ntuple, to apply the note's vertex and same-sector cuts.
- The comparison made on the extracted σ_U, σ_{TT}, σ_{LT}, as in Sec. 8.

A Reproducibility

Branch `rc-method-comparison` of `github.com/vkubarovsky/exclurad_py`, directory `docs/rc_method_pi0/`; scripts `production/analysis/rc_method_pi0*.py`. Ntuples and per-event η tables: `/Volumes/wd_14tb/rc_method_`